**ANSWER KEY**

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · A3 · STAYING HEALTHY

# Healthy habits and germs

★☆☆
Grade 5

**Learning goals**

- Recall what we need to stay healthy.
- Link diseases to how they spread.

## 1 Fill in the gaps. [6]

WORD BANK

**vaccine hygiene television exercise sugar sleep germs water**

To stay healthy we need clean air and safe water, good hygiene, nutritious food, exercise and enough sleep. Tiny living things that cause disease are called germs. A vaccine trains the body to fight a germ.

## 2 Sort the diseases by how they spread. [4]

SORT THESE

**dengue flu typhoid cholera malaria diarrhoea common cold tuberculosis**

| Dirty water or food | Air (coughs, sneezes) | Mosquito bites |
|---|---|---|
| cholera | flu | dengue |
| typhoid | tuberculosis | malaria |
| diarrhoea | common cold | |

## 3 True or false? [5]

| Statement | True | False | If false, correct it |
|---|---|---|---|
| 1. All germs cause disease. | | ✔ | Most germs are harmless; some are useful. |
| 2. Washing hands with soap for 20 seconds removes most germs. | ✔ | | |
| 3. Diabetes spreads from person to person. | | ✔ | Diabetes is not infectious; it is linked to lifestyle and genes. |
| 4. Children aged 10 need about 9 to 12 hours of sleep. | ✔ | | |
| 5. Mosquitoes breed in standing water. | ✔ | | |

**4** **Put the handwashing steps in order.** [3]

| | |
|---|---|
| 6 | Dry with a clean cloth. |
| 2 | Add soap. |
| 1 | Wet your hands with clean water. |
| 5 | Rinse well. |
| 3 | Rub palms, backs of hands and between fingers. |
| 4 | Scrub nails and thumbs. |

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · A3 · STAYING HEALTHY

# Causes, risks and choices

★★☆
Grade 5

**Learning goals**

- Link unhealthy conditions to diseases.
- Suggest behaviours that prevent disease.

**1** **For each problem, name a disease it could cause and a way to prevent it.** [8]

| Problem | Possible disease | How to prevent it |
|---|---|---|
| drinking pond water | diarrhoea / cholera | drink boiled, filtered or tube-well water |
| flower tubs full of old rainwater | dengue | empty and clean the tubs every week |
| eating without washing hands | diarrhoea, worms | wash hands with soap before eating |
| sitting with screens all day, eating crisps and sweets | obesity, diabetes later | exercise every day; eat a balanced diet |

**2** **Explain why a vaccine protects not only you but also other people.** [2]

If you cannot catch the disease, you cannot pass it on; when many people are vaccinated the germ cannot spread easily, protecting babies and sick people too.

**3** **A child has diarrhoea and is losing a lot of water. Why is ORS more helpful than plain water?** [2]

ORS replaces the salts the body has lost as well as water, and the sugar helps the body absorb the salts and water.

**4** **Rafi says, "I slept for 5 hours, but I'm fine." Explain why sleep matters for children.** [2]

Sleep helps the body grow and repair, helps the brain learn and remember, and helps the body fight infection.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · A3 · STAYING HEALTHY

# How well do we wash our hands?

★★☆
Grade 5

**Learning goals**

- Plan a fair test using a model of germs.
- Present results in a bar chart.
- Draw a conclusion and evaluate the method.

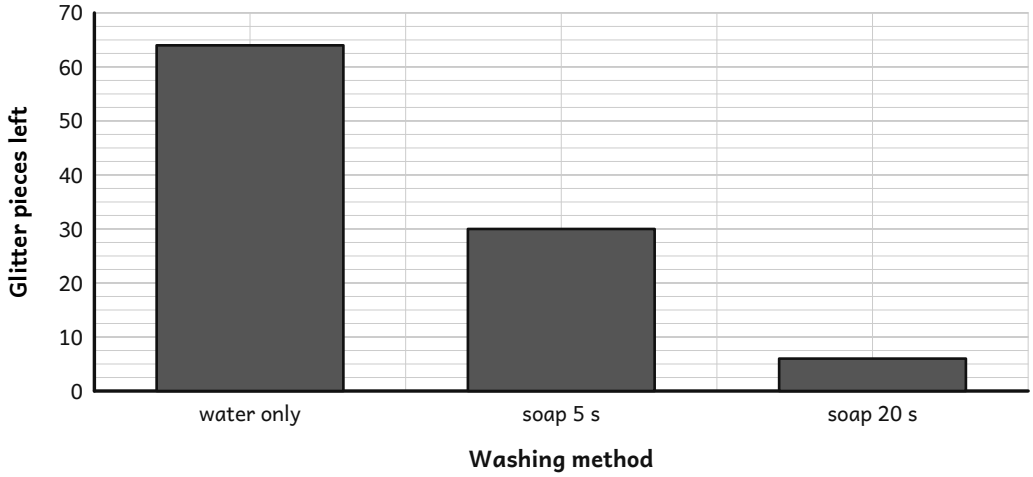
Germs are too small to see, so Class 5C used a **model**: they rubbed a teaspoon of cooking oil mixed with glitter onto their hands. The glitter stood for germs. Then they washed in different ways and counted the glitter pieces left on one hand.

**1** **Name the variables.** [3]

| | |
|---|---|
| What we change | the way of washing (water only / soap 5 s / soap 20 s) |
| What we measure | the number of glitter pieces left on the hand |
| What we keep the same | amount of oil and glitter, water temperature, same hand, same person counting |

**2** **Here are the results (average of 3 learners for each method).** [4]

| Method | water only | soap, 5 seconds | soap, 20 seconds |
|---|---|---|---|
| Glitter left | 64 | 30 | 6 |




**3** **Write a conclusion.** [2]

Washing with soap removes more "germs" than water alone, and washing for 20 seconds works best.

**4** **How is the glitter model like real germs, and how is it different?** [2]

**a)** Like real germs:

it sticks to hands and spreads to things we touch; soap helps remove it

**b)** Different from real germs:

glitter is much bigger and visible; it is not alive and cannot multiply

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · A3 · STAYING HEALTHY

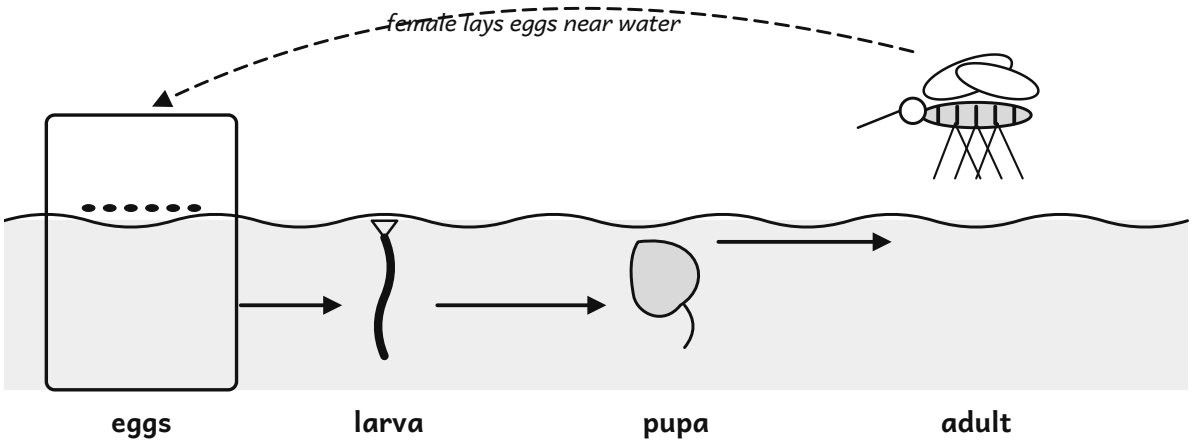
# Dengue: a city fights a mosquito

★★★
Grade 5

**Learning goals**

- Explain how a disease spreads using a life cycle.
- Interpret data about cases over the year.
- Argue for actions that reduce risk.

Science | Health | Maths | Geography | Citizenship




*Life cycle of the Aedes mosquito. From egg to adult takes about 7 to 10 days in warm weather.*

**Dengue** is a disease caused by a virus. It is spread by the bite of the female **Aedes** mosquito, which has black and white stripes and often bites in the daytime. Symptoms include a high fever, headache and pain in the muscles and joints. Most people recover with rest and fluids, but some become seriously ill and need hospital care.

Aedes mosquitoes lay their eggs in small amounts of **clean, still water**: flower tubs, old tyres, plastic containers, rooftop gardens and building sites. The eggs can survive dry for months and hatch when it rains.

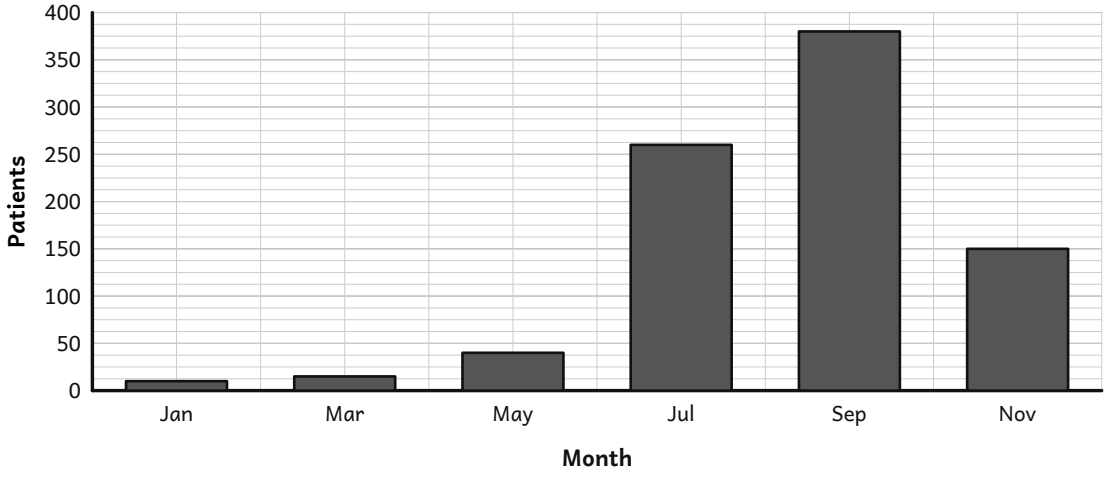
Dengue cases in Bangladesh rise during and after the monsoon. **2023 was the worst dengue year on record in the country.** City corporations spray to kill mosquitoes and run "clean-up" campaigns, but experts say every household must help by removing breeding places.

Information: Directorate General of Health Services (DGHS) and WHO dengue fact sheets.

**1** **Use the life cycle to explain why emptying water containers every week stops mosquitoes.** [2]

Mosquitoes need water to grow from egg to larva to pupa (about 7–10 days). Emptying the water every week kills the young stages before they become adults.

**2** **This chart shows example numbers of dengue patients in one city hospital each month.** [5]




| Month | Jan | Mar | May | Jul | Sep | Nov |
|---|---|---|---|---|---|---|
| Patients | 10 | 15 | 40 | 260 | 380 | 150 |

**a)** Draw the bars on the chart.

**b)** In which months are there most cases? Suggest why.

July to September (monsoon): lots of rain leaves water in containers where mosquitoes breed.

**c)** How many more patients were there in September than in May?

380 – 40 = 340

**3** **Who is responsible for stopping dengue? Give one job for each.** [3]

| Who | What they can do |
|---|---|
| each family | empty water containers weekly; use nets and repellent; cover water tanks |
| schools | check the grounds and roof for standing water; teach learners |
| the city | clean drains, remove rubbish, spray, inspect building sites |

**4** **Design a short poster message (no more than 12 words) to persuade your neighbours to act.** [2]

Your message

e.g. "Empty it, cover it, clean it: no water, no dengue!"

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · A3 · STAYING HEALTHY

# Show what you know

★★★
Grade 5

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

**1** **Find the key words.**

| K | D | E | N | G | U | E | J | K | T | M |
|---|---|---|---|---|---|---|---|---|---|---|
| P | V | E | X | E | R | C | I | S | E | V |
| K | G | D | H | J | F | S | U | S | W | J |
| C | F | S | W | Y | G | A | U | L | V | G |
| S | Y | L | D | A | G | W | M | J | I | E |
| O | J | E | V | G | T | I | D | O | R | R |
| A | G | E | K | O | M | E | E | B | U | M |
| P | S | P | Y | D | I | D | R | N | S | V |
| D | G | G | K | F | Y | E | P | B | E | P |
| U | R | I | L | U | P | V | T | B | D | P |
| V | A | C | C | I | N | E | E | U | Y | P |

**dengue** **exercise** **germ** **hygiene** **sleep** **soap** **vaccine** **virus** **water**

**2** **Which disease is spread by mosquitoes?** [1]

- ○ cholera
- ● **dengue**
- ○ flu
- ○ typhoid

**3** **Which is the best way to make pond water safer to drink?** [1]

- ○ let it stand for an hour
- ● **boil it**
- ○ add sugar
- ○ put it in the fridge

**4** **Choose one disease you learned about. Write its cause, one risk factor and how to prevent it.** [4]

| Disease | e.g. cholera |
|---|---|
| Cause | bacteria in dirty water or food |
| Risk factor | drinking unsafe water; poor sanitation |
| Prevention | safe water, latrines, handwashing, vaccine |

**5** **Challenge. Explain why a city with good drains, clean water and toilets has fewer cases of diarrhoea.** [2]

Diarrhoea germs spread through dirty water and human waste. Good drains, toilets and clean water keep waste away from drinking water and food.